10-^{7} meter (green) to 5 × 10-^{9} meter (X-ray). And similarly, the outer disk of the quasar 3C273, which you know emits infrared radiation of wavelength 5×10^{-5} meter, appears to be shining with green 5×10^{-7} meter light.

After thoroughly recording the details of the overhead spot, you turn your attention to the interior of your starship. You half expect that here, so near the hole's horizon, the laws of physics will be changed in some way, and those changes will affect your own physiology. But no. You look at your first mate, Kares; she appears normal. You look at your second mate, Bret; he appears normal. You touch each other; you feel normal. You drink a glass of water; aside from the effects of the 10-g acceleration, the water goes down normally. Kares turns on an argon ion laser; it produces the same brilliant green light as ever. Bret pulses a ruby laser on and then off, and measures the time it takes for the pulse of light to travel from the laser to a mirror and back; from his measurement he computes the speed of the light's travel. The result is absolutely the same as in an Earth-based laboratory: 299,792 kilometers per second.

Everything in the starship is normal, absolutely the same as if the ship had been resting on the surface of a massive planet with 10-g gravity. If you did not look outside the starship and see the bizarre spot overhead and the engulfing blackness all around, you would not know that you were very near the horizon of a black hole rather than safely on the surface of a planet —or you almost wouldn't know. The hole curves spacetime inside your starship as well as outside, and with sufficiently accurate instruments, you can detect the curvature; for example, by its tidal stretch between your head and your feet. But whereas the curvature is enormously important on the scale of the horizon's 300-trillion-kilometer circumference, its effects are minuscule on the scale of your 1-kilometer starship; the curvature-produced tidal force between one end of the starship and the other is just one-hundredth of a trillionth of an Earth gravity ($10^{-14}\ g$), and between your own head and feet it is a thousand times smaller than that!

To pursue this remarkable normality further, Bret drops from the starship a capsule containing a pulsed-laser-and-mirror instrument for measuring the speed of light. As the capsule plunges toward the horizon, the instrument measures the speed with which light pulses travel from the laser in the capsule's nose to the mirror in its tail and back. A computer in